Lecture 12

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Midterm 1

- Reminder that Midterm 1 will be on Wednesday, October 14 in class.
- Midterm 1 will cover:
  - Chapters 1-5 and 8-12 in the text, and
  - the material in Lectures 1-8.
- The R material covered in lab will not be on Midterm 1 but is fair game for Midterm 2 and the final.

Midterm 1 (continued)

- You can find example questions for the exam from:
  - Sample midterms posted on Canvas
  - Homework questions
  - Additional questions in the text
  - Review questions
  - Questions discussed in lecture
- You will want to focus more on solving problems and less on memorizing the lectures.

Bayes' Rule Question

Suppose that 1% of car batteries are defective, 50% of cars with defective car batteries do not start, and 0.1% of cars without defective car batteries do not start. Suppose that a car does not start.

1. What are the prior odds of a defective battery for the car?
2. What are the data odds of a defective battery for the car?
3. What are the posterior odds of a defective battery for the car?
4. What is the probability that the battery is defective given that the car does not start?

Lecture 12

1. Reading Summary - Chapter 15
2. Orderings, Permutations, and Combinations
3. Main Point for Combinations
4. Properties of Combinations
5. The Binomial Distribution
6. The Hypergeometric Distribution
7. Random Variables

Finish reading Chapters 16-17 for next time.

Reading Summary - Chapter 15

- Section 1: Introduction
  - An example of a binomial problem, step-by-step
- Section 2: The Binomial Formula
  - The general formula for binomial problems
  - Examples when the formula does not apply

Orderings, Permutations, and Combinations

Ordering a Set

- Suppose we have a set $\{a_1, a_2, a_3, ..., a_n\}$ of n objects.
- How many different ways can we order the n objects?
- Suppose we have a set $\{a_1, a_2, a_3\}$ of 3 objects (so $n = 3$).
- How many different ways can we order the 3 objects?

Ordering a Set with 3 Objects

- Answer: 6 :
  - (a_1, a_2, a_3),
  - (a_1, a_3, a_2),
  - (a_2, a_1, a_3),
  - (a_2, a_3, a_1),
  - (a_3, a_1, a_2),
  - (a_3, a_2, a_1).

Ordering a Set with 4 Objects

- Another example: suppose we have a set $\{a_1, a_2, a_3, a_4\}$ of 4 objects (so $n = 4$).
- How many different ways can we order the 4 objects?
- Answer: 24
- The 24 different ways are:
  - (a_1, a_2, a_3, a_4), (a_1, a_2, a_4, a_3),
  - (a_1, a_3, a_2, a_4), (a_1, a_3, a_4, a_2),
  - (a_1, a_4, a_2, a_3), (a_1, a_4, a_3, a_2),
  - (a_2, a_1, a_3, a_4), (a_2, a_1, a_4, a_3),

Ordering a Set with 4 Objects (Continued)

- (a_2, a_3, a_1, a_4), (a_2, a_3, a_4, a_1),(a_2, a_4, a_1, a_3), (a_2, a_4, a_3, a_1),
- (a_3, a_2, a_1, a_4), (a_3, a_2, a_4, a_1), (a_3, a_1, a_2, a_4), (a_3, a_1, a_4, a_2),
- (a_3, a_4, a_2, a_1), (a_3, a_4, a_1, a_2), (a_4, a_2, a_3, a_1), (a_4, a_2, a_1, a_3),
- (a_4, a_3, a_2, a_1), (a_4, a_3, a_1, a_2), (a_4, a_1, a_2, a_3), (a_4, a_1, a_3, a_2).

The General Rule

- For any n, the answer is:
  - $n! = n \times (n-1) \times (n-2) \times ... \times 2 \times 1$.
  - We read the value $n!$ as "n factorial."
- For example,
  - $3! = 3 \times 2 \times 1 = 6$, and
  - $4! = 4 \times 3 \times 2 \times 1 = 24$

Justification for the General Rule

- There are n options for the first object.
- There are then $n-1$ remaining options for the second object.
- There are then $n-2$ remaining options for the third object.
- ...
- There is then 1 remaining option for the last object.
- So there are a total of $n \times (n-1) \times (n-2) \times ... \times 2 \times 1$ possibilities.

Permutations

- How many ways are there to select and k objects from the previous set of n objects order matters?
- For example, suppose $n = 5$ and $k = 2$.

We have a set of 5 objects:

$$\{a_1, a_2, a_3, a_4, a_5\}$$

How many different ways can we select 2 objects from 5 where order matters?
- These are called **permutations**.

Permutations of 2 Objects from 5

- Answer: 20
  - (a_1, a_2), (a_2, a_1), (a_1, a_3), (a_3, a_1),
  - (a_1, a_4), (a_4, a_1), (a_1, a_5), (a_5, a_1),
  - (a_2, a_3), (a_3, a_2), (a_2, a_4), (a_4, a_2),
  - (a_2, a_5), (a_5, a_2), (a_3, a_4), (a_4, a_3),
  - (a_3, a_5), (a_5, a_3), (a_4, a_5), (a_5, a_4).

General Rule for Permutations

- For any n and any k, the answer is:
  - $\frac{n!}{(n-k)!} = n \times (n-1) \times (n-2) \times ...(n-k+1)$.
- For example,
  - $\frac{5!}{(5-2)!} = \frac{5!}{3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1}$
  - $= 5 \times 4 = 20$.

Justification for the Permutations Formula

- Why does this work?
  - There are n options for the first object.
  - There are then $n-1$ remaining options for the second object.
  - There are then $n-2$ remaining options for the third object.
  - ...
  - There are then $n-k+1$ remaining options for the k-th object.
- This gives $n \times (n-1) \times (n-2) \times ... \times (n-k+1)$ possibilities.

Combinations

- How many ways are there to select k objects from n objects where we do not care about order?
- How many ways are there, for example, to select 2 objects from 5 objects?
- Answer: 10
  - $\{a_1, a_2\}$, $\{a_1, a_3\}$, $\{a_1, a_4\}$, $\{a_1, a_5\}$, $\{a_2, a_3\}$,
  - $\{a_2, a_4\}$, $\{a_2, a_5\}$, $\{a_3, a_4\}$, $\{a_3, a_5\}$, $\{a_4, a_5\}$,
- These are **combinations** or subsets.

Difference between Permutations and Combinations

- Note that
  - $(a_1, a_2) \neq (a_2, a_1)$, but
  - $\{a_1, a_2\} = \{a_2, a_1\}$.
- Use parentheses for permutations, where order matters, and brackets for subsets, where order does not matter.

Combinations Formula

- In general, for any n and any k, the answer is $\binom{n}{k} = \frac{n!}{(n-k)! \times k!}$.
- We read $\binom{n}{k}$ as "n choose k."
- This the number of subsets of size k in a set of size n.
- From the previous example:
  - $\binom{5}{2} = \frac{5!}{(5-2)! \times 2!}$
  - $= \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1) \times 2!}$
  - $= \frac{5 \times 4}{2!} = 10$

Justification for Combinations Formula

- Why does this work?
- Basic idea: each combination has $k!$ associated permutations.
- For example, $\{a_2, a_5\}$ has (a_2, a_5) and (a_5, a_2).
- In general, $\binom{n}{k} = \frac{n!}{(n-k)! \times k!} = \frac{n!}{(n-k)!} \div k!$

Main Point for Combinations

- Given a set of n objects, the number of subsets of size k is equal to $\binom{n}{k} = \frac{n!}{(n-k)!\times k!} = n * (n-1) * \cdots * (n-k+1)/k!$.
- There are $\binom{n}{k}$ ways to choose k objects from n objects.
- Thus $\mathrm{combinations} = \mathrm{permutations} \div k!$,
- Or alternatively $\mathrm{permutations} = \mathrm{combinations} \times k!$

Properties of Combinations

1. $\binom{n}{k} = \binom{n}{n-k}$
   - Choosing the subset is equivalent to choosing the opposite of the subset.
2. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
   - A k element subset either contains the first element or it does not.
3. $\sum_{k=0}^{n} \binom{n}{k} = 2^n$
   - A set of size n has 2^n subsets.

- You can use property 2 to create Pascal's Triangle.

Pascal's Triangle

See Pascal's Triangle

R Code for Factorials, Permutations, and Combinations

```r
factorial(5) #5!.
```

```
## [1] 120
```

```r
#Use exp(lfactorial(n)) for large n.
#Permutations of 2 objects from 5
factorial(5)/factorial(5-2)
```

```
## [1] 20
```

```r
choose(5,2) #Combinations of 2 objects from 5
```

```
## [1] 10
```

The Binomial Distribution

The Binomial Problem

- Suppose a box contains red balls and green balls.
  - The fraction of green balls is p.
- Suppose that you following the procedure below.
  1. Pull out a ball at random.
  2. Check the color of the ball.
  3. Replace the ball in the box.
- You repeat this entire procedure n times.
  - What is the probability of drawing a green ball exactly k times?

A Binomial Example

- Suppose that the box has 70% green balls.
- Suppose that we repeat the procedure 5 times and want to know the chance of getting exactly 2 green balls.
- What is the chance of GGRRR?
- What is the chance of RGRGR?
- What is the chance of any combination that gives 2 green balls and 3 red balls?
- How many combinations are there of 2 green balls from the 5 total balls?
- What is the chance of getting exactly 2 green balls?

The Binomial Formula

- The probability of choosing exactly k green balls is:

$$\binom{n}{k} p^k (1-p)^{n-k}.$$

- The $p^k(1-p)^{n-k}$ term represents the probability of any combination of k green balls and $n-k$ red balls.
- The $\binom{n}{k}$ term represents the number of ways of choosing k green balls from n balls.

A Binomial Experiment

- The previous example was a **binomial experiment**.
- A binomial experiment must satisfy the following conditions:
  1. The outcome is **binary**, meaning that it only has two possible values.
  2. The probability of the value of interest is the same for each repetition of the experiment.
  3. Different repetitions of the experiment are independent.
- The binomial formula only applies to binomial experiments.
  - What conditions would be violated if we did not put the ball back in the box?

R Code for Binomial Probabilities

```r
#The probability of getting exactly 2 greens
#in 5 repetitions with p = 0.7
dbinom(2,5,0.7)
```

```
## [1] 0.1323
```

```r
#The probability of getting 4 or fewer greens
pbinom(4,5,0.7)
```

```
## [1] 0.83193
```

```r
#The probability of getting exactly 5 greens
dbinom(5,5,0.7)
```

```
## [1] 0.16807
```

The Hypergeometric Distribution

The Hypergeometric Problem

- Suppose a box contains r red balls and g green balls.
- You pull out a ball at random and check the color.
- You repeat this procedure n times, but unlike in the binomial problem, **you do not replace the balls after you pull them out.**
- What is the probability of drawing a green ball exactly k times?

Hypergeometric Example

- Suppose that there are 6 red balls and 10 green balls.
- We pick 5 balls and want to know the chance choosing exactly 2 red balls.
- How many ways are there of choosing 5 balls from the box?
  - Is each way equally likely?

Hypergeometric Example (Continued)

- How many ways are there of choosing 2 red balls from the box?
- How many ways are there of choosing $5 - 2 = 3$ green balls from the box?
- How many ways are there of choosing 2 red balls and 3 green balls from the box?
- What is the probability of choosing exactly 2 red balls and 3 green balls from the box?

General Solution to Hypergeometric Problem

- The probability of choosing exactly k green balls and $n-k$ red balls is:

$$\frac{\binom{g}{k}\binom{r}{n-k}}{\binom{g+r}{n}}$$

.

- The $\binom{g}{k}$ term refers to the green balls.
- The $\binom{r}{n-k}$ term refers to the red balls.
- The $\binom{g+r}{n}$ term refers to the total number of possible choices.

Using the Hypergeometric Distribution to Look for Association

- Suppose that we are investigating the association between age and smoking and we collect the following data:

Smoking/Age	18-30	Over 30
Smoker	5	20
Non-smoker	20	30

- Does this data suggest that there is an association between age and smoking?

Association between Age and Smoking

- Suppose that there is no association between age and smoking.
- We can then think of a universe of 25 smokers and 50 non-smokers.
  - The 18-30 year olds form a random sample of 25 individuals from this universe.
  - What is the chance that this random sample of 25 individuals will consist of exactly 5 smokers and 20 non-smokers?
  - What is the chance that this random sample of 25 individuals would have 5 or fewer smokers and 20 or more smokers?

Answers Using the Hypergeometric Distribution

```r
#Chance of drawing exactly 5 smokers
#25 smokers, 50 non-smokers, sample 25
dhyper(5,25,50,25)
```

```
## [1] 0.04761446
```

```r
#Chance of drawing at most 5 smokers
phyper(5,25,50,25)
```

```
## [1] 0.06836793
```

Interpretation

- We see a chance of roughly 5% of having exactly 5 smokers among the sample of 25 18-30 year olds.
- We see a chance of roughly 7% of having 5 or fewer smokers.
- This is unusual, but not that unusual.
- We will discuss the interpretation of these probabilities in more detail when we discuss hypothesis testing.
- This is a special case of **Fisher's Exact Test**.

Review Questions

- See Review_questions12.pdf

Random Variables

Variables

- Random variables can seem very confusing at first.
- It helps to step back and think about the meaning of variables in mathematics more generally.

Variable in Mathematics

Example	Comments
$4x + 2 = 10$	x "unknown," but can we can solve for it.
$x^2 = 9$	x "unknown," but we can partially solve for it.
$c = 299,792,458$	We know c.
$t = \text{Number of trees}$	We don't know t, but it has a definite value
$\sum_{i=1}^{5} i^2$	i stands for many numbers at once
$f(x) = x^2$	x can be any number we choose
$D = \text{the number on a die}$	D can be one of several possible values. Each value has a probability.

Random Variables

- A **random variable** is just a letter, often a capital letter, that stands for the outcome of a chance process.
- A random variable can take on any one of several **possible values**.
- Each of the possible values has a probability.

Example: Rolling a die

- Suppose you roll a die, and let X denote the number that comes up.
- X is a random variable.
  - The possible values of X are $1, 2, 3, 4, 5, 6$.
  - The probability that X equals 1 is $1/6$.
  - The probability that X equals 2 is $1/6$, and so on.
- Notation: $P(X = 1)$ means the probability that X equals 1.
  - Thus, for example, $P(X = 1) = 1/6$.
- Note: $P(...)$ doesn't represent a function. It just means the probability of.

Example: Drawing a ticket from a box

- Suppose you draw a ticket at random from a box with seven tickets: $\{2, 5, 5, 7, 7, 7, 9\}$
- You let X denote the number that comes up.
- X is a random variable with possible values $2, 5, 7, 9$.

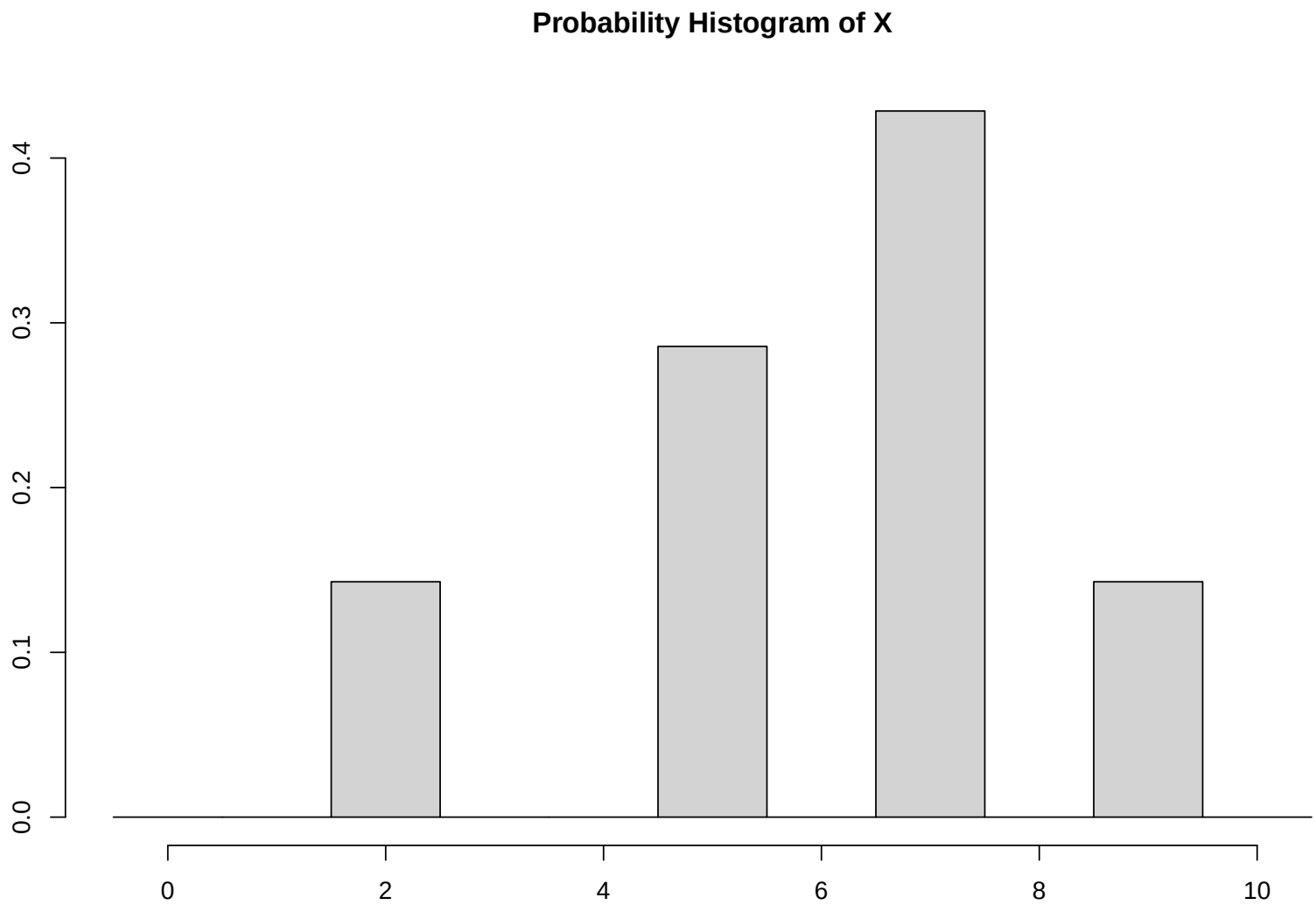
Possible Value	Probability
2	1/7
5	2/7
7	3/7
9	1/7

Probability Histograms

- We can use histograms to represent the distribution of a random variable.
- The area of each bar represents the probability that the random variable will be in a certain range.
- The key difference with data histograms:
  - a probability histogram shows probabilities for a **single random variable**, whereas
  - a data histogram shows fractions of numbers in a list.

Probability Histograms - Example

- Suppose you draw a ticket at random from a box with seven tickets: $\{2, 5, 5, 7, 7, 7, 9\}$.
- You let X denote the number that comes up.



R Random Variable Implementation

```
#Creates a sample of n independent random variables
#With distribution from the previous slide
sample_x = function(n) {
  #Create n random numbers between 0 and 1
  p = runif(n)
  #The component of the vector below is 2 if p < 1/7,
  #5 if 1/7 <= p < 3/7, 7 if 3/7 <= p < 6/7, and
  #9 if if p >= 6/7
  return(2*(p<1/7) + 5*(p>= 1/7 & p<3/7) +
         7*(p>=3/7 & p<6/7) + 9*(p>=6/7))
}
```